

MEASURES OF AGREEMENT WITH MULTIPLE RATERS: FRÉCHET VARIANCES AND INFERENCE

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ABSTRACT

Most measures of agreement are chance-corrected. They differ in three dimensions: their definition of chance agreement, their choice of disagreement function, and how they handle multiple raters. Chance agreement is usually defined following either Cohen's kappa or Fleiss's kappa. The disagreement function is usually a nominal, quadratic, or absolute value function. But how to handle multiple raters is contentious, with the main contenders being Fleiss's kappa, Conger's kappa, and Hubert's kappa, the variant of Fleiss's kappa where agreement is only said to occur if every rater agrees. More generally, multi-rater agreement coefficients can be defined in a g -wise way, where the disagreement weighting function uses g of the raters. This paper contains two main contributions. (a) We propose to use Fréchet variances to handle the case of multiple raters. The Fréchet variances are intuitive disagreement measures and turn out to generalize the nominal, quadratic, and absolute value functions to the case of more than two raters. (b) We derive the limit theory of g -wise weighted agreement coefficients, with chance agreement either of the Cohen type or Fleiss type, for the case where every item is rated by the same number of raters. Trying out three confidence interval constructions, we end up recommending calculating confidence intervals using the arcsine transform or the Fisher transform.

- (i) **Keywords:** agreement, interrater reliability, AC1, Cohen kappa
- (ii) **Running title:** Inference for measures of agreement with multiple raters
- (iii) **Data availability:** Data sharing not applicable – no new data generated